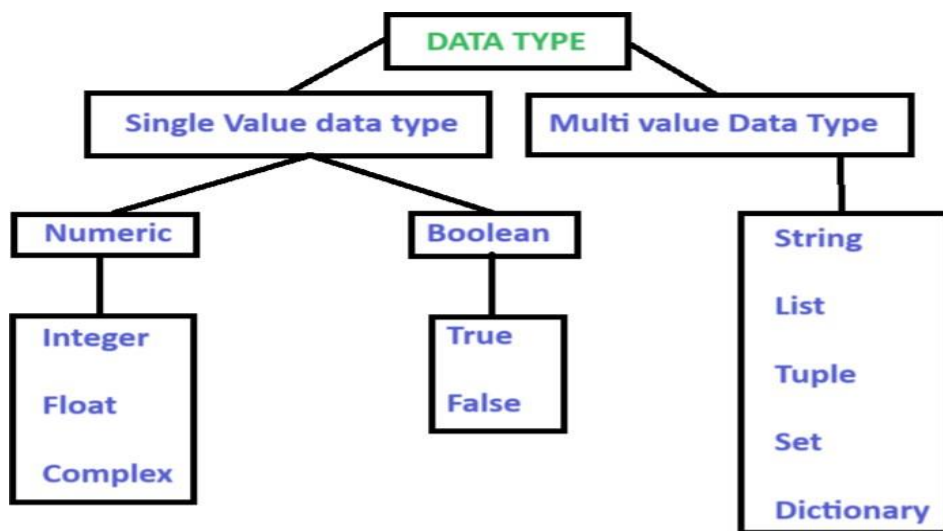


Data Types :

- Data type is used to specify the type and size of the data stored inside the variable.
- Based on the size of the value, data type got classified into two categories.

- 1) **Single Value Data type** : It is a data type where we are going to store a single value into a single variable
Ex: a = 10
- 2) **Multi value / Collection Data type**: It is a data type where we are going to store multiple values into a single variable.
Ex: a = 10, 20, 30, ...



▪ Integer (int) :

- Integer is a real number without decimal point
- Integer can be positive or Negative.
- Each and Every data type will be having 2 types of values that is, Default value and Non-Default value.
 - **Default value** : It is an initial value and it which will be internally equal to false.
Ex : 0 ----> Default value
False -----> Default value
 - **Non - Default value**: Other than default value all the values are considered as non-default value, which is internally equals to true.
Ex: Except '0' all are non-default value
- Default value for integer is '0' (zero).

🔗 **type()** : type() is inbuilt / generic function which is used to get the type of the value stored inside the variable.

Syntax : type(var_name/value)

▪ **Float (float) :**

- Float is a real number with decimal point.
- Float can be either +ve or -ve number.
- Default value for float is 0.0.

▪ **Complex (complex) :**

- Complex is a number which is a combination of real and imaginary terms **or** The number which is in the form **$a+/-bj$** is called as complex number.
- In complex number we can't use any other alphabet characters as imaginary number except 'j' or 'J'.
- If we use 'J' internally it will get converted into 'j'.
- In complex number we can't rearrange the value of 'b' and 'j'.

Ex : $a + bj$ (valid)

$a + jb$ (Invalid)

- The Default value for complex number is '0j'

▪ **Boolean (bool) :**

- Boolean is a data type which consists of only 2 values those are 'True' and 'False'.
- True is internally considered as integer '1' and False is considered as integer '0'
- Since there are only 2 values, 'True' considered as non-default value and 'False' is considered as default value.
- Boolean values are used in 2 scenarios,

- As a value while creating variable

Ex: `>>> a = True`

`b = False`

- As a Resultant while checking the condition

